Are Prediction Markets Sharp?

Kalshi, Polymarket and the Betfair Exchange Closing Line in European Football

A Registered Report

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Pre-analysis plan tagged `pap-v1`, 2026-10-25 19:22:02 UTC

Abstract

Sport has become the largest category on the new event-contract exchanges, yet existing studies judge these markets only against realised outcomes. I compare them instead with the benchmark sports bettors have long used: the Betfair Exchange closing line. Across 18 European football leagues in 2025–26, I match 3,788 Kalshi and 4,204 Polymarket match-winner markets to fixtures and compare their kickoff probabilities with Betfair's closing odds, match by match. In this exploratory season the paired log-score difference is −0.0003 nats per match for Kalshi (s.e. 0.0004) and −0.0005 for Polymarket (s.e. 0.0006). Both are statistically indistinguishable from Betfair, while the ordinary bookmaker average is clearly worse (−0.0022, s.e. 0.0005). Read at kickoff, both venues' probabilities are slightly more extreme than Betfair's. Most of that gap disappears when the venues are read 30–60 minutes earlier, so it reflects late information rather than a pricing bias. The paper's confirmatory hypothesis was registered before any of its data existed: on 2026–27 matches played after the plan was tagged, Kalshi's probabilities are within ±0.002 nats per match of the Betfair close, tested by two one-sided tests. That result will be added after the window closes on 30 June 2027. Nothing in this paper is betting or investment advice.

1 Introduction

Since mid-2025, most of the volume on Kalshi, the CFTC-regulated event-contract exchange, and on Polymarket, its crypto-settled rival, has been sport (Pew Research Center, 2026). That raises a plain question with real money behind it: do these markets price a football match as well as the markets that have priced sport for decades?

The calibration literature that has grown up around the new venues doesn't answer it. Bürgi et al. (2025) show that Kalshi's prices are fairly accurate, become more accurate as markets approach their close, and display a favourite–longshot bias. Cardozo and Rivero-Wildemauwe (2026) find the bias across 588 million Polymarket trades, except in sport. Moshrefi (2026) and Le (2026) study calibration by horizon and domain. Every one of these studies compares prices with realised frequencies. None compares them with the *closing line*, the last price of the sharpest pre-existing market and the standard benchmark in sports betting. Diercks et al. (2026) make exactly this kind of benchmark comparison for Kalshi's macroeconomic contracts against fed funds futures and surveys. This paper does it for sport.

The design is paired: two venues price the same outcome of the same match at the same moment. Most match-level noise cancels, and every season brings thousands of independent matches. The precision that results is very high. In the 2025–26 exploratory season the standard error of Kalshi's

mean log-score difference from Betfair is 0.0004 nats per match. That is small enough to detect a gap a third the size of the one separating ordinary bookmakers from the exchange.

The paper is a registered report. Everything in Section 5 comes from 2025–26, was looked at freely, and is exploratory. The single confirmatory hypothesis (Section 6) was fixed in a pre-analysis plan before any of its data existed. Its sample is every 2026–27 match kicking off after the plan's git tag. Its analysis runs once, through a script whose code is bound to that tag. Section 7 says what will be reported there.

2 Related work

Wolfers and Zitzewitz (2004) survey prediction markets as forecasting tools. The favourite–longshot bias (longshots overpriced, favourites underpriced) is one of the oldest regularities in betting markets. Snowberg and Wolfers (2010) attribute it to misperception of small probabilities rather than to risk-love. On the new venues, Bürgi et al. (2025) document it on Kalshi across contract types, with takers losing about 32% on average and makers about 10%. Cardozo and Rivero-Wildemauwe (2026) find a two-sided pattern on Polymarket in crypto and politics that is absent in sport, and show that its sign depends on how related contracts are grouped. Moshrefi (2026) reports that Kalshi's sports calibration becomes step-like in the final minutes, and Le (2026) finds horizon-dependent bias in Polymarket sports beyond a week. Across venues, Ng et al. (2025) find two-way price discovery between Polymarket and Kalshi, with Kalshi leading in sport, and Gebele and Matthes (2026) document law-of-one-price deviations of 2–4% across ten venues. Kanoria et al. (2025) estimate that about a quarter of Polymarket's reported volume is wash trading, which matters here because volume is the natural liquidity measure.

The methods are standard:

- log and Brier scores, which are strictly proper (Gneiting and Raftery, 2007);
- paired score differences, compared as in Diebold and Mariano (1995);
- equivalence tested by two one-sided tests (Schuirmann, 1987; Lakens, 2017);
- cluster-robust standard errors (Cameron and Miller, 2015);
- bookmaker margins removed proportionally or by Shin (1993)'s method;
- information content tested with a logarithmic opinion pool (Genest and Zidek, 1986).

3 Data

Fixtures and the closing line. Results and closing 1X2 odds come from Football-Data.co.uk (Football-Data.co.uk): the 18 divisions in Table 1, 6,555 league fixtures in 2025–26. Betfair Exchange closing odds are available for 6,171 of them, and for the 2026–27 season. Pinnacle closing odds cover only about 40% of 2025–26 and none of 2026–27, so Pinnacle appears only as a yardstick. Kickoff times are converted from UK time to UTC.

Kalshi. The public API lists 3,855 settled match events in these leagues for 2025–26, each with three markets (home, away, tie). For each market I read the minute candles over the three hours before kickoff. The price is the mid of the closing yes-bid and yes-ask in the last candle at or before kickoff. When the book is one-sided or wider than ten cents, the price falls back to the last trade;

that happened for 1 fixture. At kickoff the median spread is 1.3 cents and the median volume is 131,404 contracts per match.

Polymarket. The Gamma API lists 4,292 closed full-time-result events. Each has two "Will X win on DATE?" markets and one draw market; halftime, spread and other side markets are excluded. The price is the last point of the CLOB price history at or before kickoff.

Matching. Each venue event is matched by normalised team-name similarity to a football-data fixture in the same division within a day, subject to a minimum score and a margin over the runner-up. 3,788 Kalshi and 4,204 Polymarket events match (98%). Fixtures where the venue's settlement disagrees with football-data's result are dropped (4 for Kalshi, 6 for Polymarket). Every source's three prices are divided by their sum.

Table 1: Fixtures priced, 2025–26, by division

Division	Betfair close	Kalshi	Polymarket
B1	290	214	0
D1	288	306	303
D2	285	117	185
E0	358	379	380
E1	516	409	434
E2	519	46	91
EC	539	0	77
F1	287	302	302
F2	287	0	181
G1	223	119	0
I1	359	379	378
I2	359	143	261
N1	287	213	268
P1	285	221	244
SC0	216	145	150
SP1	349	379	373
SP2	437	188	295
T1	287	216	276

4 Measures and estimators

For fixture i with result $r_i \in \{H, D, A\}$, the paired log-score difference between a venue v and Betfair is

$$d_i = \log p_{v,i}(r_i) - \log p_{\mathrm{BF},i}(r_i), \tag{1}$$

which is positive when the venue was more accurate. Brier-score differences are defined with the sign reversed, so a positive value still favours the venue. Standard errors of means are CR1 cluster-robust by division × ISO week, because matches on the same weekend in the same league share shocks. In practice, clustering changes the standard error very little.

Calibration (the favourite–longshot bias) is measured by the slope b in $\operatorname{logit}\Pr(y = 1) = a + b\operatorname{logit}(p)$ over the three outcome rows of each match. $b > 1$ means favourites are underpriced and longshots overpriced. Venue and Betfair slopes are fitted jointly on the same rows, so their difference has a clustered standard error.

Information content is tested with the pool $\Pr(o) \propto \exp\{\beta_{\mathrm{BF}} \log p_{\mathrm{BF}}(o) + \beta_v \log p_v(o)\}$, fitted by maximum likelihood. $\beta_v = 0$ means Betfair encompasses the venue.

5 Exploratory evidence, 2025–26

Everything in this section is exploratory. It was looked at freely, and it informed the design in Section 6.

5.1 Accuracy

Table 2 reports the paired comparison. Kalshi and Polymarket are statistically indistinguishable from Betfair on both scores. The bookmaker average scores −0.0022 nats per match against Betfair, and Pinnacle −0.0018: both are less accurate than the exchange. That ordering is expected. It also gives the scale for what follows: the gap between ordinary bookmakers and the exchange is about 0.002 nats per match.

Table 2: Accuracy against the Betfair Exchange closing line, 2025–26 (exploratory)

			Log score			Brier score	
Source	Matches	Clusters	Mean d	s.e.	MDE	Mean	s.e.
Kalshi	3,426	353	−0.0003	0.0004	0.0012	−0.0002	0.0003
Polymarket	3,851	389	−0.0005	0.0006	0.0017	−0.0003	0.0004
Pinnacle	2,634	267	−0.0018	0.0011	0.0031	−0.0010	0.0007
Bookmaker average	6,170	609	−0.0022	0.0005	0.0015	−0.0012	0.0003

Each source is paired with Betfair on the fixtures both priced. $d > 0$ means the source was more accurate than Betfair. s.e. clustered by division × ISO week. MDE: minimum detectable effect, 5% two-sided, 80% power.

Figure 1 shows why: on the same outcome rows, Kalshi's and Betfair's calibration curves lie on top of each other.

Table 3 runs the confirmatory test of Section 6 through the confirmatory code on the whole exploratory season, with its robustness variants. In 2025–26, Kalshi's 90% interval, [−0.0010, +0.0004], lies inside ±0.002. So does Polymarket's, and so does every variant except the reads taken an hour or more before kickoff, which are noisier against Betfair's close. The bookmaker average fails, as it should.

5.2 Information content

The opinion pool (Table 4) can't separate Kalshi from Betfair. Their log probabilities are so nearly collinear that the weights trade off one for one, and the minimum detectable weight on Kalshi exceeds 1. The test is uninformative for Kalshi, so it isn't pursued; for Polymarket it is borderline.

5.3 Timing: the finding of the exploratory season

Read at kickoff, both venues' calibration slopes are slightly *lower* than Betfair's on the same rows: by −0.018 for Kalshi and −0.012 for Polymarket. Their probabilities are a little more extreme. The natural explanation is timing. The venues are read in the last minute, Betfair's "closing" snapshot is taken some time before kickoff, and the last half hour brings team news.

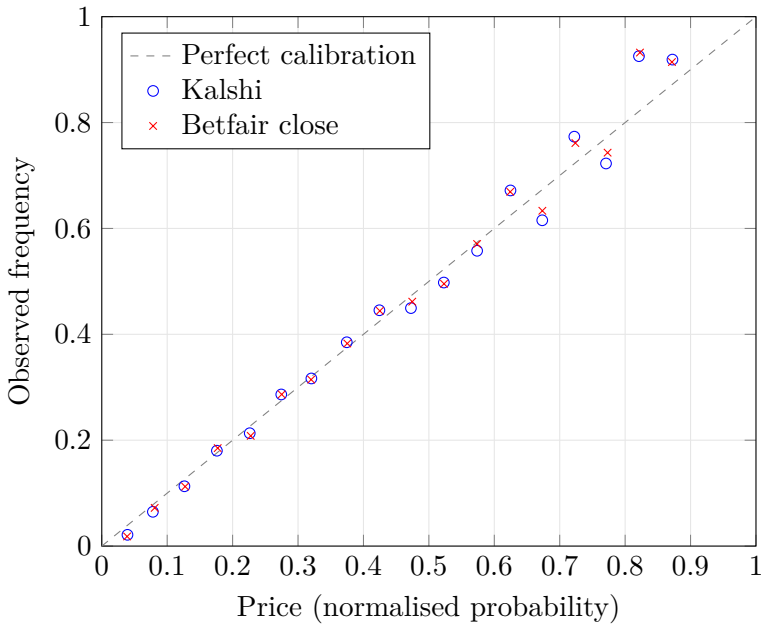


Figure 1: Calibration of Kalshi at kickoff and of the Betfair close on the same outcome rows, 2025–26, in twenty price bins (bins with at least 30 rows).

Table 3: Equivalence within ±0.002 nats per match, whole 2025–26 season (exploratory)

Specification	Matches	Mean d	s.e.	90% CI	Equivalent
H1 specification: Kalshi, log score	3,426	−0.0003	0.0004	[−0.0010, +0.0004]	yes
Polymarket (H2 specification)	3,851	−0.0005	0.0006	[−0.0015, +0.0005]	yes
Kalshi, Brier score	3,426	−0.0002	0.0003	[−0.0006, +0.0003]	yes
Kalshi read 30 min before kickoff	3,418	−0.0003	0.0006	[−0.0012, +0.0006]	yes
Kalshi read 60 min before kickoff	3,396	−0.0010	0.0008	[−0.0024, +0.0003]	no
Kalshi read 120 min before kickoff	3,258	−0.0009	0.0009	[−0.0024, +0.0006]	no
Kalshi, Shin de-vigging	3,426	−0.0004	0.0005	[−0.0011, +0.0004]	yes
Kalshi, top five leagues	1,634	−0.0001	0.0005	[−0.0009, +0.0008]	yes
Kalshi, clusters by division × date	3,426	−0.0003	0.0004	[−0.0010, +0.0005]	yes
Bookmaker average (yardstick)	6,170	−0.0022	0.0005	[−0.0031, −0.0014]	no

Two one-sided tests at 5% each: equivalence holds iff the 90% CI lies inside $(-0.002, 0.002)$. Computed with `sharpmkt.confirm`, the confirmatory script, on the exploratory season.

Table 4: Logarithmic opinion pool with Betfair, 2025–26 (exploratory)

Venue	Matches	β_{BF} (s.e.)	β_v (s.e.)	MDE for β_v
Kalshi	3,426	+1.07 (0.67)	−0.04 (0.66)	1.84
Polymarket	3,851	+0.84 (0.20)	+0.22 (0.19)	0.53

Reading the venues earlier, from the same candles, tests that explanation (Table 5, Figure 2). Polymarket's gap is gone 30–60 minutes before kickoff (−0.001 at −60 minutes). Kalshi's shrinks by about 40% but persists (−0.011, s.e. 0.005, at −60 minutes). Accuracy barely moves with the read time.

The timing pattern is the result I report from this season: *the last half hour before kickoff makes prediction-market prices more extreme than the exchange's close.* The persistent part of Kalshi's gap is smaller than one season can detect at a timing-matched read (minimum detectable gap about 0.014). It is reported as a pattern, not tested.

Table 5: Reading the venues earlier, 2025–26 (exploratory)

	Minutes before		Slope gap to Betfair		Log score d	
Venue	kickoff	Matches	Gap	s.e.	Mean	s.e.
Kalshi	0	3,255	−0.018	0.003	−0.0004	0.0004
Kalshi	30	3,255	−0.010	0.003	−0.0003	0.0006
Kalshi	60	3,255	−0.011	0.005	−0.0009	0.0008
Kalshi	120	3,255	−0.013	0.005	−0.0011	0.0009
Polymarket	0	3,850	−0.012	0.003	−0.0005	0.0006
Polymarket	30	3,850	−0.004	0.003	+0.0000	0.0006
Polymarket	60	3,850	−0.001	0.005	+0.0000	0.0008
Polymarket	120	3,850	−0.001	0.005	−0.0004	0.0008

The same fixtures at every read time within each venue. Gap: the venue's calibration slope minus Betfair's, on the same outcome rows.

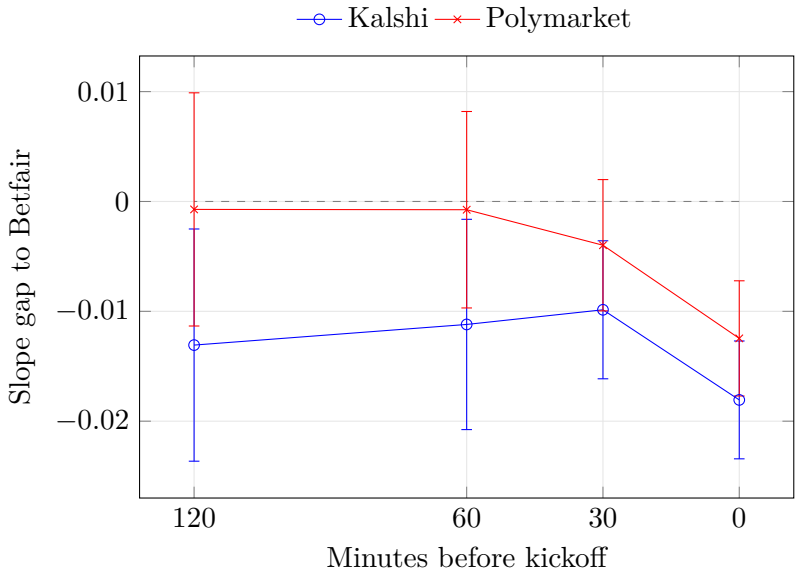


Figure 2: Calibration-slope gap to the Betfair close by read time, with 95% intervals. Negative means the venue's probabilities are more extreme than Betfair's.

5.4 Returns by price and liquidity

Table 6 shows the classic favourite–longshot shape in returns. Before fees, outcomes priced under 10% lose about a quarter to a third of the stake, but there are only a few hundred such rows and the

standard errors are wide. Table 7 finds no detectable relation between accuracy and venue volume, and no difference between the five largest leagues and the rest.

Table 6: Return to a unit stake at the venue's normalised price, by price bucket (exploratory)

	Kalshi		Polymarket	
Price	Return (s.e.)	Rows	Return (s.e.)	Rows
< 0.10	-0.234 (0.181)	325	-0.348 (0.155)	346
0.10–0.25	-0.039 (0.036)	2,832	-0.049 (0.035)	3,071
0.25–0.50	$+0.023$ (0.009)	5,459	$+0.016$ (0.008)	6,359
0.50–0.75	-0.014 (0.021)	1,433	-0.007 (0.021)	1,557
≥ 0.75	$+0.011$ (0.031)	229	$+0.033$ (0.031)	220

Before fees and the bid–ask spread. Not a trading result.

Table 7: Accuracy gap on liquidity and league tier, 2025–26 (exploratory)

Venue	Matches	Per s.d. of log volume (s.e.)	Top five minus rest (s.e.)
Kalshi	3,426	-0.0007 (0.0006)	$+0.0011$ (0.0010)
Polymarket	3,516	$+0.0005$ (0.0011)	$+0.0012$ (0.0016)

6 Pre-registered confirmatory design

The pre-analysis plan (`docs/pap/pre-analysis-plan.md`) was signed and frozen as the annotated git tag `pap-v1` at 2026-10-25 19:22:02 UTC (T0). The repository and the tag were pushed to GitHub 84 seconds later.

Hypothesis. H1: at kickoff, Kalshi's match probabilities are as accurate as the Betfair Exchange closing line, to within $\delta = 0.002$ nats per match. The null is non-equivalence, $|E[d]| \geq \delta$. It is tested by two one-sided tests at 5% each, with the clustered standard error of Section 4. Equivalence is declared if and only if the 90% confidence interval lies inside $(-\delta, \delta)$. The margin is about the gap between ordinary bookmakers and the exchange in 2025–26, so the claim reads: Kalshi is closer to the exchange than ordinary bookmakers are. The plan discloses that the equivalence framing was chosen after seeing 2025–26.

Sample. Every league fixture in the 18 divisions that kicks off after T0 and on or before 30 June 2027, and is priced by both Kalshi and Betfair. If fewer than 1,500 such fixtures exist, H1 is reported as exploratory. Polymarket (H2) is secondary, as are the robustness variants of Table 3.

Single run. The script `sharpmkt.confirm` runs once, after the window closes. It refuses to run unless `pap-v1` is an annotated tag that HEAD descends from and every file under `src/`, plus `pyproject.toml` and `uv.lock`, equals its blob at the tag. It then:

- creates an attempt marker by exclusive creation;
- pulls the 2026–27 data itself, which is the first read of the confirmatory outcomes;

- keeps the fixtures inside the window;
- writes its output once.

Every pipeline module refuses the 2026–27 season unless this script started it.

Power. Table 8 gives H1's power if the true difference is zero, by the share of the season remaining after T0. About four-fifths of 2026–27 remained at signing.

Table 8: Power of the equivalence test ($\delta = 0.002$, $\alpha = 0.05$) by share of a season

Venue	100%	70%	60%	40%
Kalshi	1.00	0.97	0.94	0.78
Polymarket	0.90	0.74	0.64	0.35

Standard error from 2025–26, scaled by the square root of the share; true difference zero.

7 Confirmatory results

To be completed after 30 June 2027.

This section will report, unedited from `docs/results/confirmatory.md`:

- the window's fixture counts;
- H1's mean, standard error, 90% interval and TOST p-value, and whether equivalence holds;
- H2 and the robustness variants;
- the exploratory timing, calibration, returns and liquidity analyses, labelled exploratory.

Any deviation from the plan will be listed with its date and reason.

8 Discussion and limitations

On the exploratory evidence, the new prediction markets price European football about as well as the sharpest exchange, and better than ordinary bookmakers. That is a strong result for venues a few years old, and the mechanism may be simple. If their market makers quote off the exchange, the venues will match it almost by construction; the collinearity in Table 4 is consistent with that. Being as sharp as the exchange is not the same as adding information to it. This design can show the first, and the data can't yet show the second.

Several limits apply:

- football-data's closing snapshot isn't time-stamped to the minute, which is why the timing analysis matters;

- coverage differs by venue and grew during the season;
- Polymarket's volume is partly wash trading;
- venue settlement occasionally disagrees with the official result;
- the confirmatory test concerns one season of one sport.

Statements

Data and code. `https://github.com/ajaiupadhyaya/sharp-markets`, tag `pap-v1`, holds all the code, the pre-analysis plan, the confirmatory script and a manifest (URL and SHA-256) of every data file and API response used. Raw data aren't redistributed; `make all` rebuilds every result from the public sources. This paper's numbers are generated by `paper/make_inputs.py`.

Preregistration. The plan was frozen as the annotated tag `pap-v1` at 2026-10-25 19:22:02 UTC and pushed to GitHub at 19:23:26 UTC, before any confirmatory fixture had kicked off.

Use of AI. The code, analysis and drafts were produced with Claude (Anthropic), running in Claude Code under the author's direction. The author set the questions, made every design decision in the plan and signed it.

Interests. The author holds no accounts or positions on Kalshi, Polymarket or Betfair and will not trade on them during the confirmatory window. The author has no funding from, or relationship with, any of them or any bookmaker.

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